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Multiyear evaluation of fruit quality diversity, stability, and climatic drivers across a diverse *Actinidia* germplasm collection

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Abstract

Understanding the genetic diversity and environmental stability of fruit quality is essential for the selection of elite kiwifruit germplasm and the development of climate-adaptive cultivars. In this study, 47 kiwifruit (*Actinidia* spp.) accessions were evaluated over four consecutive years (2021–2024) to characterize fruit-quality diversity and stability and to identify meteorological factors associated with quality variation. 8 quality traits—fruit weight (FW), dry matter (DM), soluble solids content of ripe fruit (SSC_R), sugar, titratable acidity (TA), SSC_R/TA and Sugar/TA ratios, and vitamin C (Vc)—were analyzed using descriptive statistics, correlation, multivariate analysis, and stability coefficients. Associations between quality traits and meteorological factors during key developmental windows were evaluated using Spearman's rank correlation analysis. The results revealed substantial phenotypic diversity: Vc exhibited the greatest variation (CV = 100.75%), whereas DM was the most stable trait (CV = 8.87%), and a strong predictor of sweetness of ripe fruit ($R^2 = 0.66$ with SSC_R). Genetic background significantly influenced trait expression, with *A. eriantha* hybrids showing exceptionally high levels of Vc, whereas *A. rufa* hybrids exhibiting small fruit size and low levels of Vc. Multivariate analysis classified the 47 accessions into 4 clades with distinct quality profiles. Stability analysis indicated that environmental variation among years was the primary driver of total phenotypic fluctuation, while certain accessions (e.g., S1, S9, S16, S18, and S43) exhibited both high quality and superior stability. Meteorological conditions during specific developmental windows were significantly associated with several quality traits. DM was positively associated with maximum temperature, diurnal temperature range, and temperature variability ($|r| \approx 0.38$ – 0.41), whereas TA and the sugar–acid ratio were associated with early-season heat accumulation, temperature variability, and preharvest rainfall ($|r| \approx 0.30$ – 0.47). These associations were stage dependent, and no single meteorological factor consistently explained fruit-quality variation. Overall, this study reveals substantial genetic diversity in kiwifruit fruit quality, identifies germplasm combining high quality with cross-year stability, and demonstrates the stage-specific associations of meteorological conditions with key quality traits, providing valuable resources for breeding high-quality, broadly adaptable cultivars and developing climate-adaptive orchard management strategies.

Keywords *Actinidia*, Fruit quality, Variation, Evaluation, Climate, Phenology

Introduction

Kiwifruit (*Actinidia* spp.), widely recognized as the “king of fruits”, is valued globally for its exceptional nutritional properties, including high levels of vitamin C (Vc), dietary fiber, antioxidants, and essential minerals, as well as its considerable economic importance (Richardson et al. 2018). In recent decades, the global kiwifruit industry

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has experienced rapid expansion. China, as the center of origin and the top producer, is dominated by both cultivated areas and production outputs worldwide (Zhong et al. 2022; Qi et al. 2020). The sustainable development of the kiwifruit industry critically relies on the breeding and widespread adoption of superior cultivars to meet increasingly diverse market demands for improved fruit appearance, flavor, nutrition, and storability (Qi et al. 2020; Zhong et al. 2022).

Actinidia germplasm resources are exceptionally diverse, encompassing widely cultivated species such as *A. chinensis* var. *chinensis* and *A. chinensis* var. *deliciosa*, along with numerous wild accessions and interspecific hybrids, such as *A. rufa* × *A. chinensis* and *A. chinensis* × *A. eriantha*. These diverse genetic resources harbor valuable traits, including high Vc content, improved disease resistance, and environmental adaptability (Qi et al. 2024; Ferguson 2007; Ferguson and Huang 2007). This rich genetic reservoir underpins the broad phenotypic variability observed in fruit quality attributes, including fruit weight (FW), dry matter (DM), flesh color, sugar–acid balance, Vc, dietary fiber, and mineral composition, which are crucial for developing cultivars that meet consumer and market preferences (Li et al. 2025). For example, distinct flesh colors are associated with specific nutritional strengths: red-fleshed kiwifruits generally accumulate higher levels of flavonoids and anthocyanins, green-fleshed varieties are characterized by abundant chlorophyll and dietary fiber, whereas yellow-fleshed types often exhibit higher carotenoid and soluble sugar contents (Li et al. 2025; Yuan et al. 2023). Moreover, recent studies have utilized comprehensive physicochemical profiling and principal component analysis (PCA) to rank cultivars such as ‘Donghong’, ‘Jinyan’, ‘Jinkui’, ‘Hayward’, and ‘Huate’ among the top performers in terms of overall quality, highlighting the potential of systematic evaluation in guiding targeted breeding (Yuan et al. 2023). However, fully exploiting *Actinidia* germplasm resources requires a deeper understanding of the stability of fruit quality traits under varying environmental conditions and cultivation practices (Qi et al. 2024).

Current research on kiwifruit fruit quality have primarily followed two approaches: multi-cultivar evaluations within a single season and location or single-cultivar analyses across multiple production regions. These studies widely applied multivariate methods such as PCA and clustering analysis to evaluate cultivars on the basis of integrated traits such as DM and soluble solids content (SSC), and some have begun linking consumer preference to objective measures or exploring soil–fruit nutrient relationships (Harker et al. 2009; Zhang et al. 2025; Chen et al. 2024; Lv et al. 2022; Wang et al. 2021). Nevertheless, several important knowledge gaps remain. First,

multiyear evaluations of fruit quality stability across large germplasm collections are still limited, restricting the ability to distinguish genetically superior and environmentally stable accessions from those strongly affected by annual environmental fluctuations (Tait et al. 2018; Li et al. 2022). Second, systematic analyses of how specific meteorological factors (e.g., temperature, precipitation, and diurnal range) during distinct key developmental windows precisely regulate final fruit quality attributes remain insufficiently explored, limiting the development of predictive models and a deeper understanding of genotype-by-environment interactions in kiwifruit production (Rajan et al. 2024).

To bridge these critical research gaps, this study systematically evaluated 47 kiwifruit accessions cultivated under consistent orchard conditions over 4 consecutive years (2021–2024). Specifically, we aimed to assess the diversity, interannual stability, and relationships among major fruit quality traits. In addition, we tried to identify germplasms with consistently superior and stable performance and determine the associations between key meteorological factors during critical developmental stages and the formation of important fruit quality attributes. By integrating multiyear stability assessment with stage-specific climate–quality correlation analysis, this study provides insights into the genetic and environmental determinants of kiwifruit quality and offers a scientific basis for precision breeding strategies, including the selection of cultivars with high DM content and stable flavor profiles, as well as climate-adaptive orchard management.

Materials and methods

Fruit materials

A total of 47 kiwifruit (*Actinidia* spp.) germplasm accessions, primarily *A. chinensis* and hybrids with *A. rufa* and *A. eriantha*, were evaluated in this study. These accessions represent the main commercial cultivars planted in China and new cultivars or selections/lines from the wild or from intraspecific and interspecific hybridization (Table 1). Representative fruit images of selected commercial cultivars and distinctive germplasms are shown in Fig. 1. All vines were cultivated in a commercially managed orchard located in Danjiangkou city, Hubei Province, China (32°48′N, 111°12′E). In 2017, the scions of all these accessions were grafted onto ‘Miliang No. 1’ rootstock of *A. chinensis* var. *deliciosa*. Each germplasm accession consisted of 3 biological replicates, with approximately 5 or 10 vines per replicate (the exact number depended on the availability of healthy and uniformly growing vines after natural loss and removal of weak plants over the four-year trial). Fruit quality evaluations were conducted over 4 consecutive growing seasons from

Table 1 List of *Actinidia* accessions used in this study

Sample code	Sample name	Genetic background	Sample code	Sample name	Genetic background
S1	Mantianhong	<i>A.eriantha</i> × <i>A.chinensis</i>	S25	ZK9	<i>A.rufa</i> × <i>A.chinensis</i>
S2	Mantianhong 2	<i>A.eriantha</i> × <i>A.chinensis</i>	S26	Ac15	<i>A.rufa</i> × <i>A.chinensis</i>
S3	Jintao	<i>A.chinensis</i> var. <i>chinensis</i>	S27	ZK8	<i>A.rufa</i> × <i>A.chinensis</i>
S4	Jinyuan	(<i>A.eriantha</i> × <i>A.chinensis</i>) × <i>A.chinensis</i>	S28	Ac16	<i>A.rufa</i> × <i>A.chinensis</i>
S5	Jinmèi	(<i>A.eriantha</i> × <i>A.chinensis</i>) × <i>A.chinensis</i>	S29	ZK6	<i>A.rufa</i> × <i>A.chinensis</i>
S6	Jinxia	<i>A.chinensis</i> var. <i>chinensis</i>	S30	Ac17	<i>A.rufa</i> × <i>A.chinensis</i>
S7	Wuzhi 3	<i>A.chinensis</i> var. <i>chinensis</i>	S31	ZK5	<i>A.rufa</i> × <i>A.chinensis</i>
S8	Cuiyu	<i>A.chinensis</i> var. <i>chinensis</i>	S32	ZK7	<i>A.rufa</i> × <i>A.chinensis</i>
S9	Huayou	<i>A.chinensis</i> var. <i>chinensis</i>	S33	Ac18	<i>A.rufa</i> × <i>A.chinensis</i>
S10	Jinzao	<i>A.chinensis</i> var. <i>chinensis</i>	S34	Acc elite line	<i>A.chinensis</i> var. <i>chinensis</i>
S11	Jinfeng	<i>A.chinensis</i> var. <i>chinensis</i>	S35	Acc elite line	<i>A.chinensis</i> var. <i>chinensis</i>
S12	Chuhong	<i>A.chinensis</i> var. <i>deliciosa</i>	S36	ZK10	<i>A.chinensis</i> × <i>A.eriantha</i>
S13	Dongmei	<i>A.chinensis</i> var. <i>deliciosa</i>	S37	Ace elite line	<i>A.chinensis</i> × <i>A.eriantha</i>
S14	Jinmèi	<i>A.chinensis</i> var. <i>deliciosa</i>	S38	ZK12	<i>A.chinensis</i> × <i>A.eriantha</i>
S15	Xuxiang	<i>A.chinensis</i> var. <i>deliciosa</i>	S39	ZK11	<i>A.chinensis</i> × <i>A.eriantha</i>
S16	Miliang 1	<i>A.chinensis</i> var. <i>deliciosa</i>	S40	Acc elite line	<i>A.chinensis</i> var. <i>deliciosa</i>
S17	Hayward	<i>A.chinensis</i> var. <i>deliciosa</i>	S41	ZK13	<i>A.chinensis</i> var. <i>deliciosa</i> × <i>A.chinensis</i> var. <i>deliciosa</i>
S18	Jinkui	<i>A.chinensis</i> var. <i>deliciosa</i>	S42	Acc elite line	<i>A.chinensis</i> var. <i>chinensis</i>
S19	ZK2	<i>A.chinensis</i> var. <i>deliciosa</i> × <i>A.chinensis</i> var. <i>deliciosa</i>	S43	Acc elite line	<i>A.chinensis</i> var. <i>chinensis</i>
S20	ZK3	<i>A.chinensis</i> var. <i>deliciosa</i> × <i>A.chinensis</i> var. <i>deliciosa</i>	S44	Jinyu	<i>A.chinensis</i> var. <i>chinensis</i>
S21	ZK1	<i>A.chinensis</i> var. <i>deliciosa</i> × <i>A.chinensis</i> var. <i>deliciosa</i>	S45	Hort16A	<i>A.chinensis</i> var. <i>chinensis</i>
S22	ZK4	<i>A.chinensis</i> var. <i>deliciosa</i> × <i>A.chinensis</i> var. <i>deliciosa</i>	S46	Jinnong	<i>A.chinensis</i> var. <i>chinensis</i>
S23	Donghong	<i>A.chinensis</i> var. <i>chinensis</i>	S47	Fengyue	<i>A.chinensis</i> var. <i>chinensis</i>
S24	Ac14	<i>A.rufa</i> × <i>A.chinensis</i>			

2021 to 2024 under the same orchard management conditions. Among the 47 accessions, fruit quality data for 36 accessions were obtained from our previous study (Shen et al. 2023).

Fruit harvest and storage

Once the fruit SSC of fruits reached approximately 7.0%, fruits were randomly harvested from 3 biological replicates. At harvest, fruit maturity indices, including FW, firmness, SSC, and DM, were immediately measured using 30 fruits per accession (10 fruits of each replicate). The remaining fruits were then transported to the laboratory of Wuhan Botanical Garden, Chinese Academy of Sciences (located in Wuhan city), where they arrived within 24 h of harvest. Healthy fruits of uniform size were packed into plastic bags and directly stored at 1–2 °C in a cold room for 8–12 weeks until they approached the softening and ripening stages. Before final quality evaluation, the fruits were transferred to 20 °C conditions and stored for up to 7 d to allow further ripening.

For final quality assessment, 20 ripe fruits per accession were selected in 2021–2022, whereas 30 ripe fruits

per accession were selected in 2023–2024. Fruits were selected based on edible firmness (approximately 4–8 N; the actual four-year average ranged from 3.7–8.3 N with a mean of 5.2 N; Table 2). SSC and firmness of ripe fruit (SSC_R and firmness_R) were first measured individually for each fruit. Subsequently, the pulp tissue without skin from every 6–7 fruits (or 10 fruits) was pooled to form 1 biological sample, leading to 3 biological replicates per accession for subsequent analyses of soluble sugar, titratable acidity (TA) and Vc contents. The tissue samples were homogenized with distilled water or oxalic acid solution (20 g/L) and stored at –25 °C until further analysis.

Fruit maturity measurement

Fruit firmness was measured using a Fruit Texture Analyzer (FTA; Güss model GS15; Gusstoday Manufacturing Ltd., South Africa) fitted with a 7.9-mm penetrometer probe. Approximately 1 mm of skin and underlying flesh was removed before measurement, and the probe was inserted into the fruit flesh at a speed of 5 mm/s to a depth of 7.9 mm after triggering at 50 g. The maximum

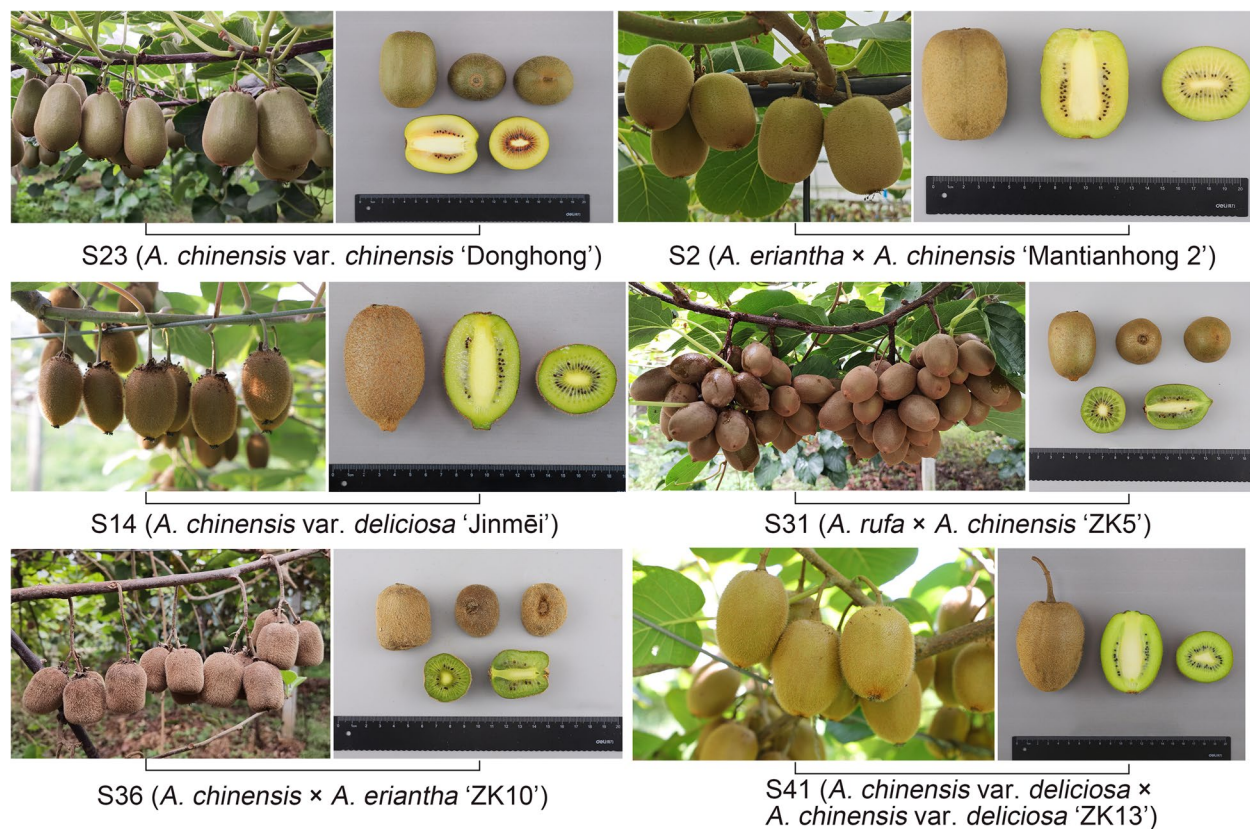


Fig. 1 Fruit morphology of selected commercial cultivars and distinctive special accessions of *Actinidia*. The sample name and genetic background are provided in Table 1. The cultivars 'Donghong', 'Mantianhong 2', and 'Jinmēi' are red-fleshed, yellow-fleshed, and green-fleshed commercial cultivars, respectively. 'ZK5', 'ZK10', and 'ZK13' represent interspecific or intraspecific hybrid germplasms of *Actinidia* with unique traits

force was recorded as the firmness value in kgf, which was converted to N, where 1 kgf=9.81 N. Firmness was measured twice at 90° separation at the equator of each fruit, and these values were averaged to obtain a fruit firmness value. Fruit SSC was determined with a digital refractometer (PAL-1; Atago, Itabashi-ku, Tokyo, Japan) on juice squeezed from the equator of each fruit. DM content was measured by drying a 2-mm-thick transverse fruit slice collected from the equator of the fruit at 65 °C for at least 24 h, and the dry weight was calculated as a percentage of the fresh weight.

Internal quality measurement

Soluble nonreducing sugars were hydrolyzed with concentrated hydrochloric acid (HCl) at 80 °C to produce reducing sugars, and the total soluble sugars were subsequently quantified by spectrophotometry using the 3,5-dinitrosalicylic acid (DNS) method according to the agricultural industrial standard method (NY/T 2742–2015) and are presented as a percentage (%) of glucose. The TA content was determined using 0.1 M sodium hydroxide (NaOH) to titrate the sample solution to a pH

of 8.2 under an automatic titrator (HI931; Hanna Instruments, Rhode Island, USA), essentially following the national standard method (GB 12456–2021), and the results are presented as percentages (%) of citric acid. The SSC_R/TA and sugar/TA ratios were calculated by dividing the SSC_R or soluble sugar content by the TA content, respectively.

The Vc content was determined using the 2,6-dichloroindophenol titrimetric method according to the national standard GB 5009.86–2016. Measurements were conducted using the same automatic titrator, and Vc content was expressed as mg/100 g fresh weight.

Comprehensive evaluation of fruit quality

A comprehensive evaluation of fruit quality was conducted by integrating multivariate analyses and ranking stability assessments on 47 kiwifruit accessions. The analysis was based on the four-year grand means of 8 core quality indices: FW, DM, SSC_R, total sugars, TA, the SSC_R/TA ratio, the sugar/TA ratio, and Vc. 3 maturity-related indices (firmness at harvest, SSC at harvest,

Table 2 The mean fruit quality for 47 *Actinidia* accessions calculated for 2021–2024

Sample code	FW (g)	FF_H (N)	SSC_H (%)	DM (%)	FF_R (N)	SSC_R (%)	Sugar (%)	TA (%)	SSC_R/TA	Sugar/TA	Vc (mg/100 g)
S1	73.74	52.78	8.39	20.14	4.69	16.74	11.67	1.13	15.15	10.56	62.16
S2	92.68	48.14	8.74	17.84	3.81	15.55	11.09	1.12	14.26	10.2	84.55
S3	112.2	61.14	9.2	17.28	5.33	14.38	9.74	0.96	14.95	10.12	119.88
S4	92.87	70.58	8.68	17.29	5.17	15.52	10.21	1.26	12.4	8.2	80.19
S5	91.95	69.02	8.16	16.53	6.88	13.89	9.76	1.67	9.98	7.05	96.69
S6	88.97	57.7	8.35	17.3	5.71	14.44	10.36	1.04	14.59	10.54	119.53
S7	91.09	70.43	8.84	18.59	5.48	14.47	9.97	0.94	15.61	10.7	179.53
S8	62.75	70.14	7.81	18.26	6.19	14.38	9.75	0.95	15.54	10.5	137.5
S9	74.28	61.48	8.59	20.11	4.29	16.69	11.37	1.01	16.73	11.32	166.01
S10	81.03	60.09	8.54	19.11	3.92	15.18	9.68	1.4	10.94	6.97	103.64
S11	95.62	62.99	8.26	15.64	5.62	11.96	6.55	1.33	9.18	5.1	100.17
S12	47.84	53.24	7.17	19.23	4.27	13.69	8.64	0.96	15.24	9.43	129.01
S13	42.12	75.34	8.07	22.36	4.11	17.09	11.73	1.09	16.7	11.41	102.89
S14	62.98	74.64	8.21	24.12	5.76	20.06	14.53	0.91	23.1	16.85	102.23
S15	67.06	93.35	7.91	19.32	4.87	16.24	9.88	1.05	15.98	9.97	104.82
S16	104.17	69.18	8.76	19.81	7.3	16.01	11.29	1.65	9.82	5.97	112.96
S17	97.05	81.07	7.74	17.9	8.29	14.21	10.27	1.21	11.93	8.65	41.47
S18	94.49	86.74	8.79	21.07	5.88	17.15	11.62	1.2	14.78	9.84	166.89
S19	75.39	66.71	7.93	18.26	6.07	14.47	9.36	1.58	9.38	5.99	111.63
S20	85.93	74.57	9.1	18.42	5.85	14.81	9.51	1.43	10.86	7	93.07
S21	111.84	62.76	7.98	15.26	5.66	12.49	7.72	1.59	8.34	5.24	95.77
S22	81.65	54.42	8.75	17.95	4.7	14.5	8.69	1.9	7.88	4.75	126.33
S23	90.98	55.5	8.69	17.2	5.19	14.27	10.57	0.85	17.42	12.95	112.59
S24	35.09	66.47	8.64	18	4.81	15.44	10.48	1.18	13.3	9.04	15.26
S25	39.18	53.95	9.87	17.91	3.81	15.58	11.12	1.02	16.01	11.65	24.24
S26	30.98	68.11	8.76	17.81	4.6	14.58	9.91	1.24	12	8.22	13.56
S27	31.83	60.95	9.9	19.34	5.48	16.51	11.36	1.08	16.44	11.39	16.5
S28	32.2	61.06	8.53	17.67	4.26	15.17	10.1	1.67	10.52	7.09	88.75
S29	39.29	58.49	9.58	16.94	4.06	14.98	10.81	0.99	15.74	11.53	14.75
S30	33.64	58.94	9.86	18.67	4.06	16.06	10.37	1.09	15.21	10.3	39.93
S31	33.78	62.6	9.31	16.72	4.54	14.36	9.38	1.2	12.27	8.05	33.95
S32	37.66	69.57	8.58	18.76	5.23	16	11.33	1.01	16.35	11.58	38.73
S33	29.82	76.92	10.08	18.27	4.85	15.93	11.57	0.85	20.81	15.41	76.94
S34	63.42	61.56	9.22	19.29	6.59	16.65	11.62	1.39	12.15	8.52	72.43
S35	48.46	71.09	8.08	18.98	6.4	14.91	10.29	1.29	12.05	8.44	77.35
S36	27.1	63.62	9.2	17.39	4.85	14.48	10.02	1.19	12.49	8.68	458.03
S37	31.44	55.5	8.9	15.39	5.09	13.11	8.64	1.14	11.66	7.77	433.01
S38	30.6	65.54	10.03	18.77	5.43	15.65	10.55	1.1	14.82	9.95	561.28
S39	28.3	45.95	9.61	16.92	3.65	14.18	9.88	1.37	10.58	7.44	513.55
S40	87.43	74.73	10.49	18.96	6.05	15.7	10.31	1.46	11.24	7.45	117.69
S41	110.93	59.15	9.46	17.18	5.32	13.71	10.49	0.99	14.36	11.11	45.7
S42	67.66	72.63	8.33	19.42	6.42	15.46	10.51	1	15.92	10.84	63.93
S43	76.25	65.72	9.06	19.47	6.07	16.39	11.04	1.41	11.95	8.03	69.85
S44	77.63	64.47	9.1	19.88	4.87	16.84	11.31	1.02	17.11	11.65	85.93
S45	69.87	72.88	7.79	18.8	4.96	15.12	9.81	0.99	16.18	10.84	76.04
S46	86.76	51.84	8.88	17.73	4.58	14.64	9.42	1.43	10.48	6.73	115.2
S47	88.83	47.95	7.75	16.91	4.25	13.42	8.66	1.28	10.63	6.86	80.03
Mean	67.17	64.72	8.76	18.39	5.22	15.17	10.28	1.20	13.64	9.32	120.90
SD	27.06	9.85	0.73	1.63	0.98	1.39	1.25	0.25	3.15	2.50	121.80
CV	40.29	15.22	8.28	8.87	18.84	9.13	12.13	20.42	23.13	26.81	100.75
Min	27.10	45.95	7.17	15.26	3.65	11.96	6.55	0.85	7.88	4.75	13.56
Max	112.20	93.35	10.49	24.12	8.29	20.06	14.53	1.90	23.10	16.85	561.28

FW, Fruit weight; FF_H, fruit firmness at harvest; SSC_H, soluble solids content at harvest; FF_R, firmness of ripe fruit; SSC_R, SSC of ripe fruit; TA, Titratable acidity; Vc, Vitamin C

and firmness_R) were excluded from the comprehensive quality assessment to focus on core quality attributes.

To explore the underlying structure of the germplasm collection, hierarchical clustering analysis (HCA) was performed. Before HCA, the trait data were standardized using Z score normalization. The analysis used Euclidean distance as the similarity metric and Ward's method as the linkage criterion to form distinct, interpretable clusters of accessions. For dimensionality reduction and the derivation of a composite quality score, PCA was employed. The same Z score standardized data were used for an initial PCA to identify major patterns of variation. For final quality evaluation and ranking, a separate PCA was subsequently conducted on the data that had undergone range normalization (0–1 scale). Because TA is considered a negative indicator of fresh fruit quality, it was inversely normalized. The PC scores from this analysis were used to calculate a composite quality score, based on which the 47 accessions were ranked from 1 (highest overall quality) to 47 (lowest overall quality).

The stability among years of the fruit quality ranking was visualized in a heatmap, which displays the annual PCA-based rank (1–47) of each accession over the four-year period. Accessions on the vertical axis were ordered in descending order of their composite quality score (derived from four-year means), with the highest-ranked accession placed at the bottom. To exemplify the multi-trait profiles underlying the ranking, a radar chart was plotted for 3 representative accessions spanning the quality spectrum: S14 (Rank 1, highest overall quality), S29 (Rank 23, intermediate quality), and S11 (Rank 47, lowest overall quality). This chart displays the range-normalized values (0–1) of all 8 quality traits for each selected accession.

Stability analysis of fruit quality based on variation coefficients

To assess fruit quality stability across traits and germplasms and explore its relationship with overall fruit quality, 3 coefficients of variation (CVs) were calculated from the four-year phenotypic data of the 8 fruit quality traits of the 47 kiwifruit accessions: FW, DM, SSC_R, sugar, TA, the SSC_R/TA ratio, the sugar/TA ratio, and Vc. CV_total was used to quantify total phenotypic variation based on all individual fruit measurements collected across the 4 years. CV_between captured between-year variation, calculated from annual means per accession. CV_year represented within-year variation, which was calculated as the average of yearly CVs from individual fruit measurements per season. For most traits, a simple arithmetic mean of the annual CVs was used to calculate CV_year. For SSC_R, for which the annual sample sizes

differed between years ($n=20$ or 30 fruits per accession), a sample-size-weighted mean of the annual CVs was used instead.

At the trait level, radar charts displayed the mean CV_total, CV_between, and CV_year of each trait across all the accessions. At the germplasm level, a scatter plot of mean CV_between versus mean CV_year was generated, with point size scaled to mean CV_total to compare stability profiles among accessions. A heatmap detailing the CV_total for all trait-accession combinations, with accessions ordered vertically by descending mean CV_total. To examine the relationship between fruit quality and interannual stability, a second scatter plot related the PCA-derived comprehensive quality rank (x-axis) to the mean CV_between (y-axis), with the point size proportional to the mean CV_year. This integrated visualization simultaneously highlights interannual stability, within-year consistency, and overall quality performance.

Climate-quality correlation analysis

Daily meteorological data, including the average (T_{avg}), maximum (T_{max}), and minimum (T_{min}) air temperatures, as well as the daily precipitation (rainfall) of the experimental site for the 4 growing seasons (2021–2024), were downloaded from the National Meteorological Information Center (<http://data.cma.cn>; Table S1). A series of biologically relevant climatic indices were calculated from these primary measurements for subsequent analysis.

For each of the 47 *Actinidia* accessions, meteorological variables were extracted for critical developmental periods in each year from 2021 to 2024. The full bloom and harvest dates for each accession across the 4 years are provided in Table S2, and these dates were used to calculate the days after anthesis (DAA) and the corresponding preharvest windows for meteorological data extraction. The target windows were the fruit development stages (30, 40, 50, and 60 DAA) and the preharvest periods (10, 20, 30, and 40 days before harvest; PH10–40). The 30–60 DAA window represents the rapid fruit enlargement phase, and the PH10–40 window represents the key DM accumulation period before harvest. 4 time points within each window were chosen to provide relatively dense temporal coverage for identifying climatic variables most strongly associated with fruit quality. For each period and each original daily index, key statistical summaries (e.g., mean, sum, and count of events) were calculated, generating a comprehensive dataset of 100 derived climatic variables (Table S2).

The following key meteorological factors were calculated from the primary daily data:

Diurnal temperature range (DTR): DTR was calculated as the difference between daily maximum and minimum temperatures: $DTR = T_{\max} - T_{\min}$.

Daily growing degree days (Daily_GDD): Daily_GDD was calculated using a base temperature (T_{base}) of 10 °C: $\text{Daily_GDD} = \max(T_{\text{avg}} - T_{\text{base}}, 0)$.

Effective growing degree days (Eff_GDD): To account for potential suppression of growth at high temperatures, an upper temperature threshold of 30 °C was incorporated:

$$\text{Eff_GDD} = \max\{\min(T_{\text{avg}}, 30) - 10, 0\}$$

Extreme event indicators: A Hot_Day was defined when $T_{\max} > 35$ °C; a Cold_Night was defined when $T_{\min} < 10$ °C; and a Rain_Day was defined when rainfall > 0.1 mm.

Temperature–humidity index (TH_Index): A simplified index was calculated as: $\text{TH_Index} = T_{\text{avg}} / (\text{Rainfall} + 1)$.

Heatwave events (Heat_Wave): A heatwave event was defined as ≥ 3 consecutive days with $T_{\max} > 35$ °C.

Effective precipitation (Eff_Rain): Effective precipitation was estimated by applying retention coefficients of 90% to rainfall ≤ 5 mm and 80% to rainfall > 5 mm to account for potential runoff.

Short-term climatic stability was assessed using the 7-d sliding window coefficient of variation for daily mean temperature (TempCV).

The statistical analysis focused on 8 key fruit quality traits: FW, DM, SSC_R, sugar content, TA, the ratios of SSC_R/TA and Sugar/TA, and Vc. Spearman's correlation analysis was performed between these 8 quality traits and the 100 derived meteorological variables across the 8 key developmental periods (30–60 DAA and PH10–40). This nonparametric method was chosen to capture monotonic relationships without assuming linearity or a normal distribution of the data. The resulting correlation matrix was visualized using a heatmap. To further illustrate the strongest associations identified by the correlation analysis, 4 representative fruit quality traits (DM, soluble sugar content, TA, and the SSC_R/TA ratio) were plotted against their respective most strongly correlated meteorological variables: 60DAA_MaxTemp, 50DAA_TempCV, 30DAA_EffectiveGDD, and 30DAA_TempCV, respectively. These scatter plots were fitted with linear regression lines, and the corresponding 95% confidence intervals of the fit are displayed to depict the central trend and its uncertainty amidst biological variability.

Statistical analysis

All data are presented as means of the corresponding biological or technical measurements, as appropriate. Pairwise correlations among the fruit quality traits were

assessed using Pearson correlation analysis, whereas associations between quality and climate was determined using Spearman correlation analysis. All statistical analyses, including HCA, PCA, and stability analysis based on CVs, climate-quality correlation and all graphical visualizations (radar charts, scatter plots, and heatmaps), were performed using Origin 2024b software (OriginLab Corporation, Northampton, MA, USA).

Results

Analysis of the fruit quality traits of 47 kiwifruit accessions

The mean values of the fruit quality traits for the 47 kiwifruit accessions from 2021 to 2024 are summarized in Table 2. Considerable phenotypic variation was observed across the germplasm collection. Specifically, the single-FW ranged from 27.10 g to 112.20 g, with an average of 67.17 g. DM content at harvest varied between 15.26% and 24.12%, with an average of 18.39%, whereas SSC_R ranged from 11.96% to 20.06%, with an average of 15.17%. Soluble sugar content varied from 6.55% to 14.53%, averaging 10.28%, while TA ranged from 0.85% to 1.90%, with an average of 1.20%. The SSC_R/TA ratio varied from 7.88 to 23.10, with an average of 13.64, and the Sugar/TA ratio ranged from 4.75 to 16.85, with an average of 9.32. Vc content exhibited the widest variation, ranging from 13.56 to 561.28 mg/100 g fresh weight, with a mean of 120.90 mg/100 g.

The extent of phenotypic variation for different accessions differed markedly among quality traits, as reflected by their CVs. The Vc content had the highest CV (100.75%), followed by the single-FW (40.29%). In contrast, DM and SSC_R were the most stable traits, both with CV values of approximately 9.0%.

From the perspective of genetic background, the variation range of fruit quality indices is associated with the characteristics of the *Actinidia* species group. For instance, the accessions with an *A. rufa* background typically presented lower fruit weights (usually 30–40 g) and lower Vc contents (mostly within 13–40 mg/100 g), whereas the accessions with an *A. eriantha* background presented extremely high Vc contents (mostly approximately 500 mg/100 g) and relatively small fruit sizes (approximately 30 g). In contrast, the accessions with *A. chinensis* var *chinensis* and *A. chinensis* var *deliciosa* backgrounds generally presented relatively greater fruit weights and moderately high Vc contents. Other quality indices, such as DM, SSC_R, sugar, TA, and their derived ratios, inherently showed relatively narrow variation ranges (Table 2).

Comprehensive evaluation of fruit quality

The Pearson correlations among the 11 fruit quality indices were analyzed on the basis of their mean values for

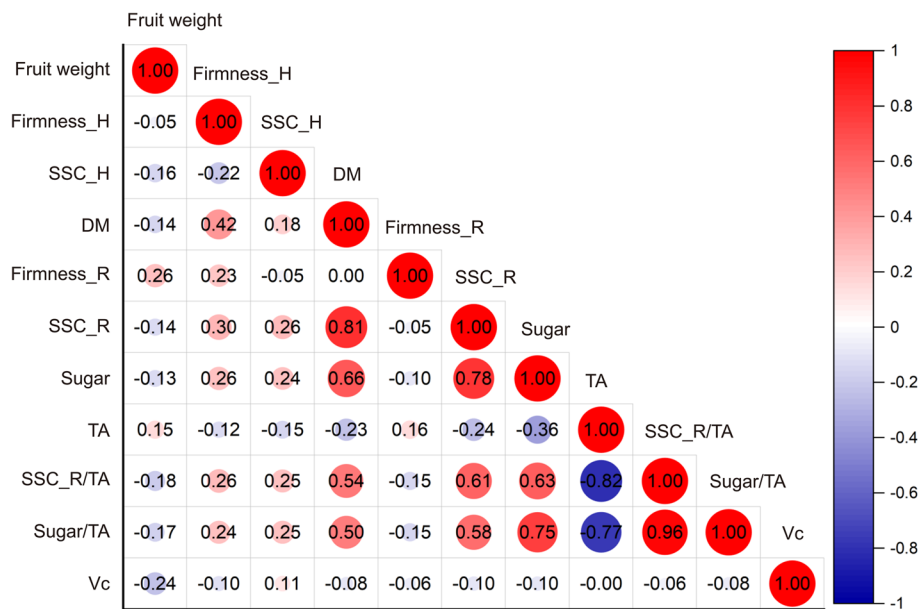


Fig. 2 Correlation analysis of the fruit quality parameters of 47 *Actinidia* accessions from 2021 to 2024 on the basis of their annual mean values. The values shown in circles represent the Pearson correlation coefficient; $n = 188$ (47 accessions $\times$ 4 years). Firmness_H and SSC_H, fruit firmness and SSC at harvest, respectively; DM, dry matter of fruit at harvest; Firmness_R and SSC_R, firmness and SSC of ripe fruit, respectively; TA, titratable acidity; SSC_R/TA, SSC_R-TA ratio; Sugar/TA, sugar-TA ratio; Vc, vitamin C

each year for 2021–2024 ($n = 188$). DM at harvest was strongly positively correlated with SSC_R and the sugar content (Pearson $r = 0.81$ and 0.66 , respectively), followed by the SSC_R/TA ratio and the Sugar/TA ratio. Moreover, the SSC_R/TA ratio and Sugar/TA ratio were strongly positively correlated with each other (Pearson $r = 0.96$). Both ratios were strongly negatively correlated with the TA content (Pearson $r = -0.82$ or -0.77) and positively correlated with the SSC_R or sugar content (Pearson $r = 0.58$ – 0.75). No significant correlations were observed among the other major fruit quality indices (Fig. 2).

To evaluate the potential of DM content at harvest as an indicator of subsequent fruit quality, linear regression models were developed between the DM and both the SSC_R and soluble sugar content using four-year annual mean data (Fig. 3). Both relationships were highly significant ($P < 0.01$). DM content at harvest showed a strong positive relationship with SSC_R ($R^2 = 0.66$) and a moderate positive relationship with soluble sugar content ($R^2 = 0.46$), supporting its potential as an early indicator of subsequent fruit sweetness. The regression model predicts that when the DM at harvest reaches or exceeds 16.8%, the corresponding SSC_R is likely to meet or surpass the common edible threshold of approximately 14.0%.

Both HCA and PCA were conducted to elucidate the underlying structure of the germplasm collection on the basis of the four-year grand means of 8 core fruit quality

traits. The HCA revealed a clear hierarchical structure, segregating the 47 accessions into 4 major clades (I–IV), each characterized by a distinct fruit quality profile (Fig. 4a). Clade I comprised accessions with relatively high DM and SSC_R contents. Clade II was primarily distinguished by fruit size and was further subdivided into Clade IIa (large fruit size with moderate Vc content) and Clade IIb (small fruit size combined with low Vc content). The latter (Clade IIb) consisted of hybrid accessions derived from *A. rufa*. Clade III was characterized by relatively high TA content. Clade IV was distinguished by exceptionally high Vc content and contained 4 hybrid accessions (S36–S39) with genetic contributions from *A. eriantha*.

The PCA results were largely in agreement with the HCA clustering results (Fig. 4b). The first 2 PCs together explained 72.83% of the total variance, with PC1 and PC2 accounting for 55.01% and 17.82%, respectively. The loading plot showed that DM, SSC_R, and soluble sugar content clustered closely, consistent with their strong positive associations. TA and FW formed another distinct group, whereas the SSC_R/TA and sugar/TA ratios were closely positioned. Vc was separated from the other quality traits, indicating that it represented a relatively independent dimension of fruit quality variation. In the corresponding score plot, the 47 accessions were distributed across 4 quadrants, and their spatial positions, reflecting their composite

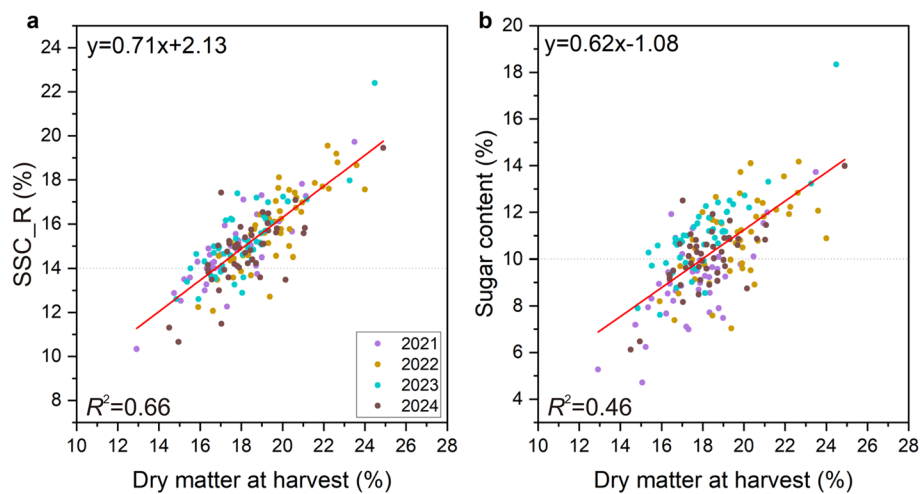


Fig. 3 Linear regression analyses between dry matter (DM) content at harvest and soluble solids content of ripe fruit (SSC_R; **a**) and sugar content (**b**) in 47 *Actinidia* accessions from 2021 to 2024. Each point represents the annual mean value of an accession for a given year. The mean values for SSC_R, sugar content, and DM content at harvest are shown for the four growing seasons (2021–2024)

quality profiles, generally agreed with their classification in the HCA dendrogram.

The stability of the comprehensive quality rankings derived from PCA was further evaluated across the four consecutive growing seasons (Fig. 5a). Accessions at the upper and lower extremes of the ranking distribution, particularly those within approximately the top 10 and bottom 10 positions, generally maintained relatively stable rankings across years. In contrast, accessions occupying the middle portion of the ranking distribution (ranks 11–37) showed greater year-to-year fluctuations. Statistical analysis of year-to-year ranking consistency yielded Spearman and Pearson correlation coefficients ranging from approximately 0.4 to 0.6 (data not shown), confirming a moderate but not perfect positive correlation across seasons.

The multitrait architecture underlying the ranking was further elucidated by analyzing 3 representative accessions spanning the quality spectrum: S14 (top-ranked, Rank 1), S29 (median-ranked, Rank 23), and S11 (bottom-ranked, Rank 47) (Fig. 5b). Their quality profiles, visualized via radar charts of range-normalized trait values (0–1), revealed distinct patterns. S14 exhibited superior performance across the six key traits, with values approaching 1.0, but showed only moderate values for FW and Vc content. In contrast, S11 exhibited relatively low normalized values for 7 traits but a comparatively high value for FW. S29 presented an intermediate profile for 6 quality traits, with notably minimized Vc and FW. These contrasting multitrait profiles illustrated the differentiation in overall fruit quality among accessions with different PCA-derived comprehensive rankings.

Stability analysis of fruit quality for *Actinidia* accessions

Analysis of the variation coefficients, within-year (CV_year), between-year (CV_between), and total (CV_total), across eight traits in 47 kiwifruit accessions revealed distinct stability patterns. For seven of the 7 traits, CV_between was greater than CV_year and was broadly comparable to CV_total. Only the SSC_R index showed comparable CV_between and CV_year values. Furthermore, both DM and SSC_R exhibited the lowest total variation (7.36% and 10.25%, respectively), indicating high trait stability, whereas the remaining 6 traits displayed substantially greater total variation, ranging from 14.74% to 28.86% (Fig. 6a).

To visualize stability patterns at the germplasm level, a scatter plot of mean CV_year versus mean CV_between was constructed on the basis of the average values of 8 fruit traits across 47 kiwifruit accessions (Fig. 6b). The results revealed 2 key patterns. First, CV_between consistently exceeded CV_year in all the accessions, indicating that interannual environmental variation is the primary driver of total phenotypic variation (CV_total). Second, the accessions were distributed across 4 quadrants delineated by the mean CV_between and CV_year. Those accessions located in quadrant I (e.g., S12, S13, S28, and S33) displayed high values for both variation components, reflecting heightened sensitivity to environmental fluctuations at both temporal scales. Conversely, the accessions in quadrant III (e.g., S2, S3, S4, S9, and S43) exhibited concurrently low CV_between and CV_year, indicating superior phenotypic stability across years.

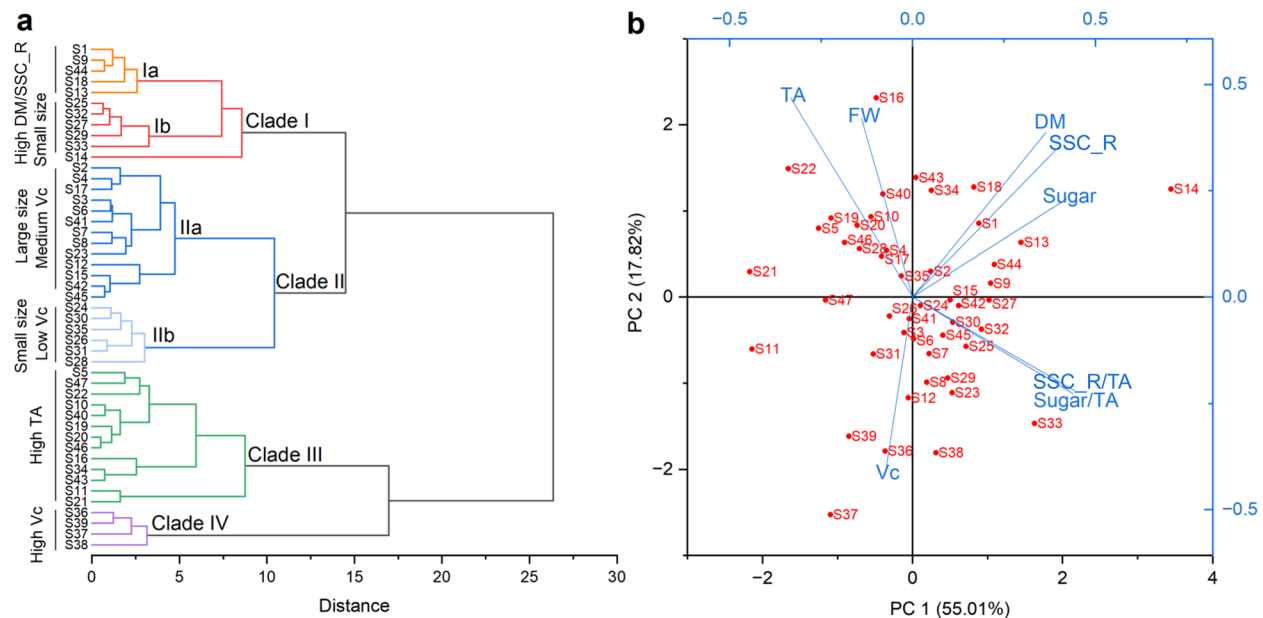


Fig. 4 Hierarchical clustering analysis (HCA; **a**) and principal component analysis (PCA; **b**) of 8 core fruit quality traits in 47 kiwifruit accessions based on their grand means across 4 years. The S1–S47 sample codes represented 47 *Actinidia* accessions. FW, fruit weight; DM, dry matter; SSC_R, soluble solids content of ripe fruit; TA, titratable acidity; SSC_R/TA, SSC_R-TA ratio; Sugar/TA, sugar-TA ratio; Vc, vitamin C

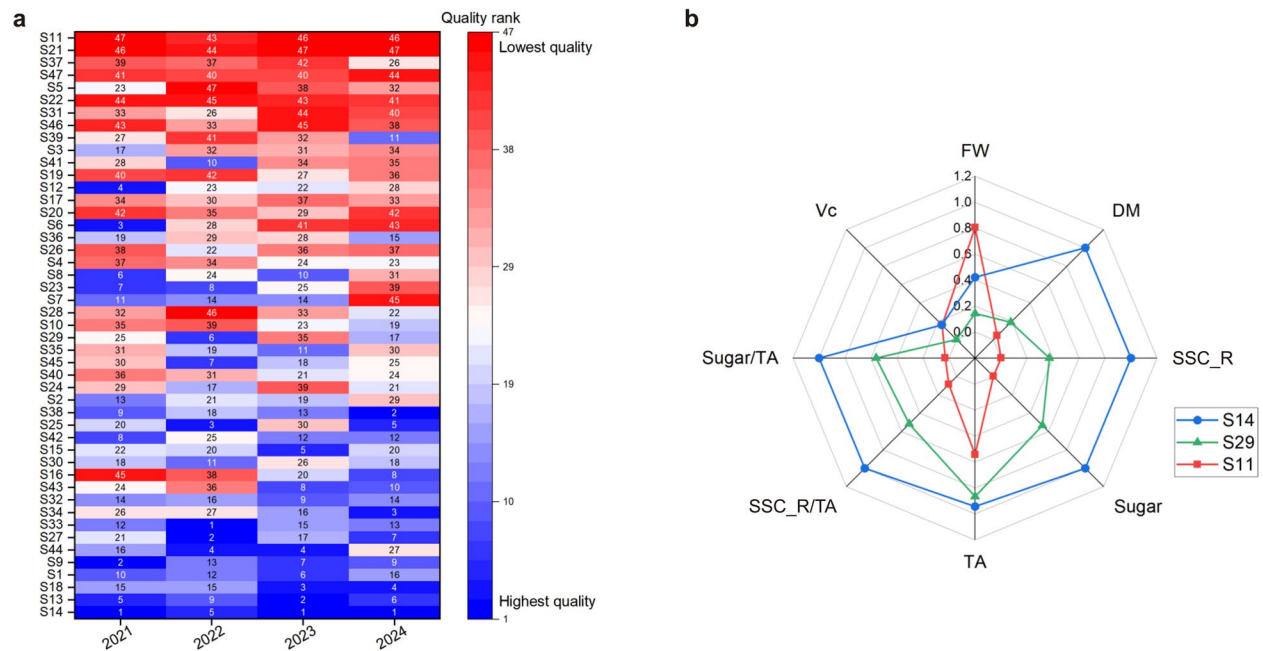


Fig. 5 Comprehensive evaluation and representative comparison of fruit quality in the *Actinidia* accessions. **a** Heatmap of comprehensive quality rankings for 47 *Actinidia* accessions across 4 years. The color scale indicates the ranking position, with dark blue representing the highest ranks (closest to 1) and dark red the lowest ranks (closest to 47). Accessions on the vertical axis are arranged from bottom to top according to their average comprehensive principal component analysis ranking derived from four-year mean data, with rank 1 representing the highest overall quality and rank 47 the lowest. **b** Comparison of quality trait profiles for 3 representative kiwifruit accessions. Accessions were selected on the basis of principal component analysis (PCA)-derived comprehensive quality rankings: S14 (top-ranked, Rank 1), S29 (median-ranked, rank 23), and S11 (bottom-ranked, Rank 47). The quality trait values represented four-year grand means normalized to a 0–1 scale using min–max transformation (TA inversely scaled). FW, fruit weight; DM, dry matter; SSC_R, soluble solids content of ripe fruit; TA, titratable acidity; SSC_R/TA, SSC_R-TA ratio; Sugar/TA, sugar-TA ratio; Vc, vitamin C

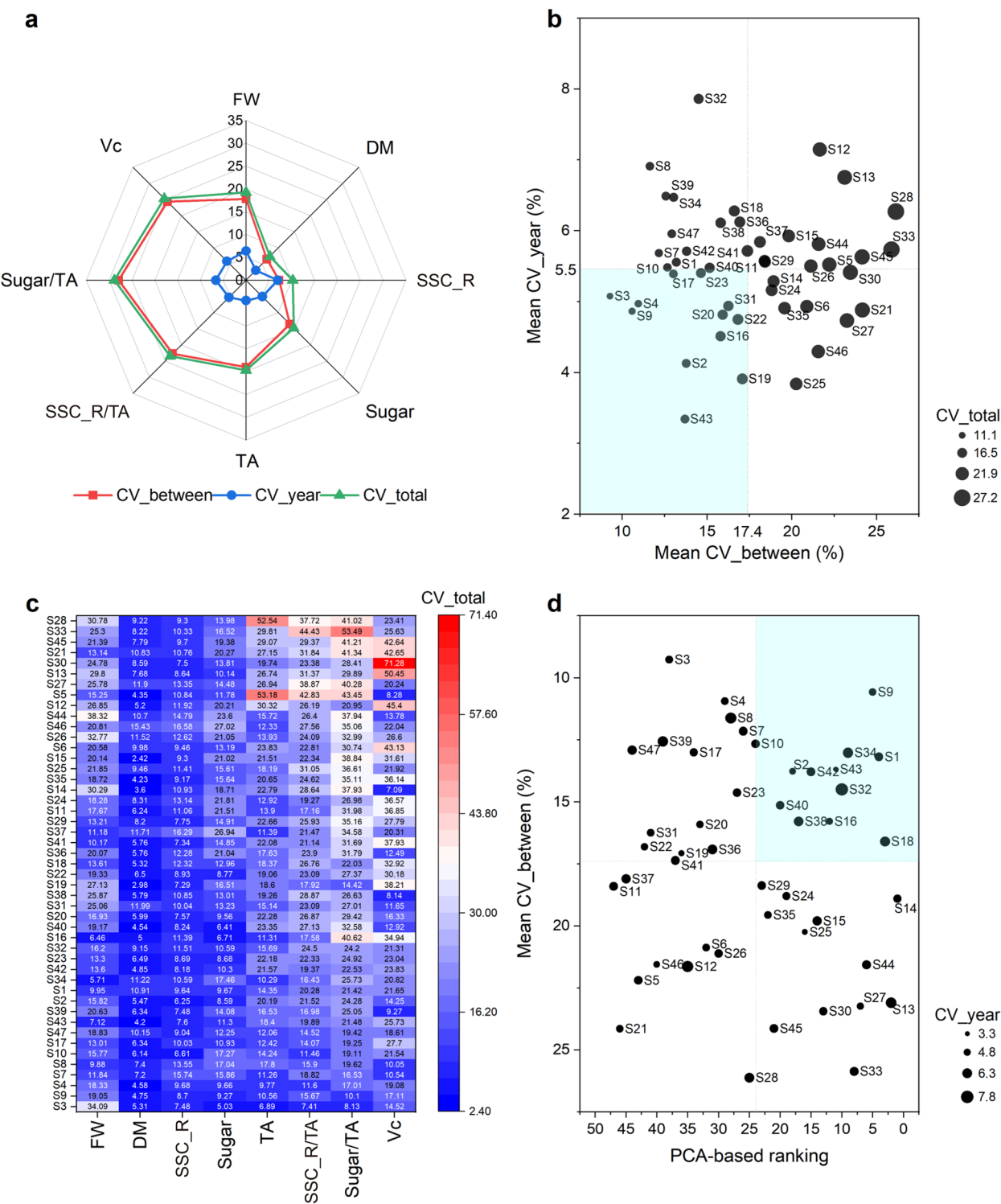


Fig. 6 Stability analysis of the fruit quality traits of 47 *Actinidia* accessions. **a** Radar chart displaying the mean coefficients of variation (CV)_{year}, CV_{between}, and CV_{total} for the 8 quality traits. FW, fruit weight; DM, dry matter; SSC_R, ripe soluble solids content of ripe fruit; TA, titratable acidity; SSC_R/TA, SSC_R–TA ratio; Sugar/TA, Sugar–TA ratio; Vc, vitamin C. **b** Scatter plot of mean CV_{year} versus mean CV_{between} the 8 quality traits for 47 accessions. Size of points is proportional to the mean CV_{total}. **c** Heatmap of the CV_{total} values for the 8 quality traits across the 47 accessions. Accessions on the vertical axis are arranged in descending order on the basis of the mean CV_{total} of the 8 quality traits (from bottom to top). **d** Scatter plot of principal component analysis (PCA)-based comprehensive quality ranking against the mean CV_{between} the 8 quality traits for 47 accessions. Size of points is proportional to the mean CV_{year}

The patterns of total phenotypic variation (CV_total) were further detailed in a heatmap (Fig. 6c). At the germ-plasm level, the mean CV_total ranged from 11.1% (S3) to 27.2% (S28), with an overall mean of 18.6%, reflecting substantial diversity in stability. At the trait level, DM and SSC_R consistently presented the lowest CV_total across all the accessions, confirming their high stability, whereas the other 6 traits exhibited markedly greater variation.

Finally, the relationship between comprehensive fruit quality (PCA-based ranking) and phenotypic stability was assessed in a scatter plot relating quality rank to mean CV_between, with point size scaled to mean CV_year (Fig. 6d). The results indicated that the 47 accessions were distributed nearly uniformly across the 4 quadrants defined by the mean quality rank and mean CV_between. Accessions in quadrant I (e.g., S1, S9, and S18) combined high comprehensive quality with low CV_between, indicating that they were elite genotypes with both superior performance and high stability. In contrast, accessions in quadrant III (e.g., S5, S12, and S21) exhibited low comprehensive quality coupled with high CV_between, indicating poor performance and high environmental sensitivity. Notably, mean CV_between and mean CV_year showed no strong correlation across the accessions, suggesting that between-year and within-year variation represented relatively distinct components of phenotypic variability.

Analysis of the associations between fruit quality traits and meteorological factors

Spearman correlation analysis was conducted to assess the relationships between 8 key fruit quality traits and meteorological variables across critical developmental windows (30–60 DAA) and preharvest periods (PH10–PH40). The analysis, encompassing 47 *Actinidia* accessions over four years, revealed distinct and significant associations for 5 of the quality traits (Fig. 7), highlighting trait-specific climatic drivers and sensitive phenological windows.

Numerous significant associations were detected using the predefined criteria of $|r| \geq 0.30$ and $P < 0.0001$. In contrast to the other quality traits, FW and SSC_R exhibited no strong correlations ($|r| \geq 0.3$) with the meteorological variables examined. Vc was an exception, with 5 instances of moderate correlations ($|r| = 0.25–0.3$). DM was positively correlated with maximum temperature, with the strongest association observed for 60DAA_MaxTemp ($r = 0.384$), as well as with the frequency of hot days and DTR. In contrast, DM was consistently negatively correlated with minimum temperature across the DAA periods, with the strongest association reaching $r = -0.310$. Sugar content was positively correlated with temperature variability (TempCV) during fruit development, with

the strongest correlation observed for 50DAA_TempCV ($r = 0.411$), but was negatively associated with mean temperature, Eff_GDD, and TH_Index, with the strongest negative correlation observed for 40DAA_EffectiveGDD ($r = -0.383$).

TA exhibited the largest number of significant associations, with 24 meteorological variables showing $|r| > 0.30$. It was positively influenced by mean temperature and effective GDD during early development (30–40 DAA, $r = 0.322–0.460$), but negatively associated with temperature variability, with the strongest relationship observed for 30DAA_TempCV ($r = -0.462$). TA was also negatively associated with total and effective precipitation during the preharvest periods, as well as with hot-day and heat-wave frequencies.

The derived quality ratios, SSC_R/TA and sugar/TA, showed some of the strongest individual associations with meteorological variables. Both ratios were positively associated with temperature variability, exemplified by the relationship between sugar/TA and 30DAA_TempCV ($r = 0.513$), as well as with total and effective precipitation during the preharvest periods. In contrast, both ratios were negatively associated with mean temperature, effective GDD, minimum temperature, and TH_Index during the DAA periods. Substantial overlap was observed in the key meteorological drivers (e.g., TempCV, early-season heat accumulation, and preharvest rainfall) for TA, SSC_R/TA, and sugar/TA, underscoring their interconnected role in shaping the sugar–acid balance.

1 variables (y-axis) for 47 kiwifruit accessions $\times$ 4 years ($n = 188$ observations). Meteorological data correspond to two critical phenological windows each year: fruit development (30–60 days after anthesis, DAA) and pre-harvest maturation (10–40 days before harvest, PH10–40). Red and blue indicate positive and negative associations, respectively, with color intensity representing the magnitude of the correlation coefficient. All correlation coefficients are shown; $|r| \geq 0.30$ was used as the threshold for strong associations, with $P < 0.0001$. Quality traits were assessed either at commercial harvest (fresh weight, FW; dry matter, DM) or after ripening (soluble solids content of ripe fruit, SSC_R; sugars; titratable acidity, TA; SSC_R/TA ratio; sugar/TA ratio; and vitamin C, Vc).

The relationships between the 4 selected quality traits and their respective most influential meteorological drivers are shown in Fig. 8. Scatter plots with linear regression fits illustrate the associations of DM with 60DAA_MaxTemp, the sugar content with 50DAA_TempCV, TA with 30DAA_EffectiveGDD, and SSC_R/TA with 30DAA_TempCV. The linear models explained a modest proportion of the variance for each trait (Pearson's $R^2 = 0.11, 0.11, 0.11$, and 0.14 , respectively). The

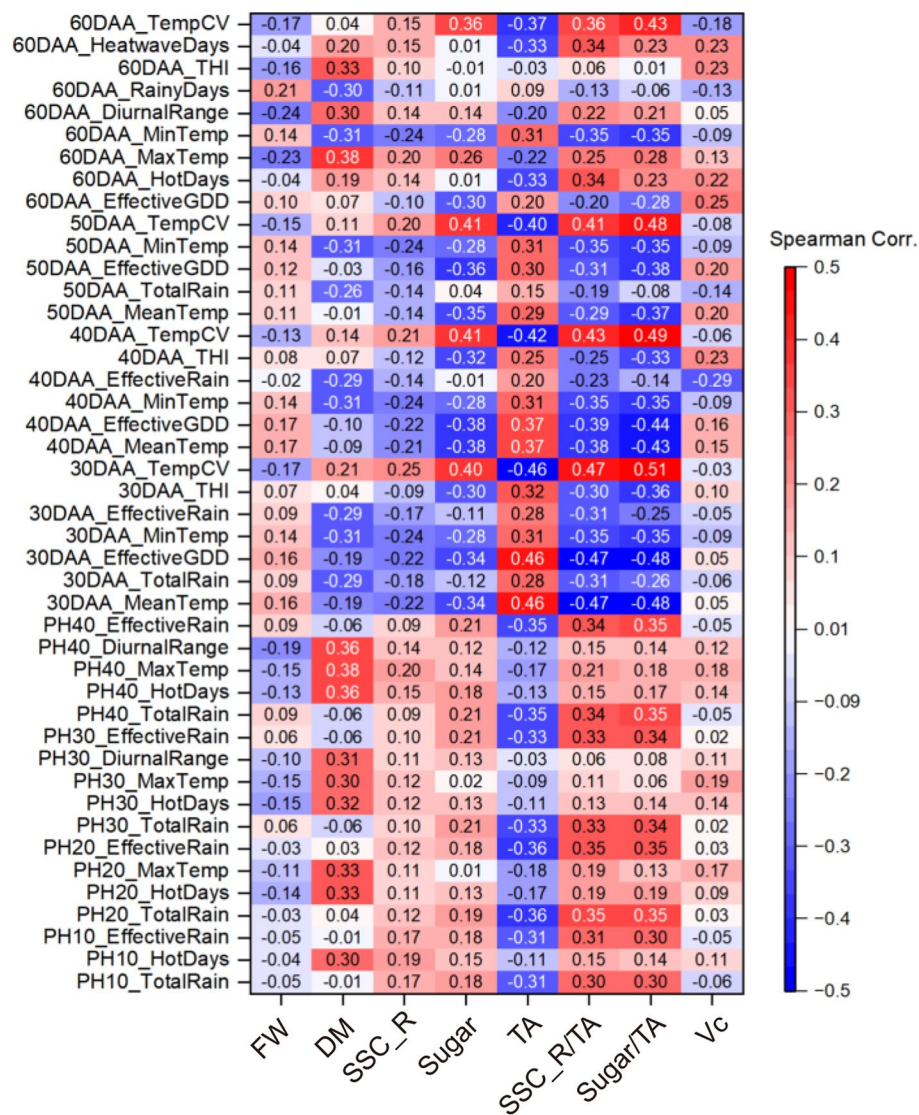


Fig. 7 Heatmap of Spearman correlation coefficients between kiwifruit quality traits and meteorological variables during key developmental periods. Correlation analysis was performed between 8 quality traits (x-axis) and meteorological variables (y-axis) for 47 kiwifruit accessions $\times$ 4 years ($n=188$ observations). Meteorological data correspond to two critical phenological windows each year: fruit development (30–60 days after anthesis, DAA) and pre harvest maturation (10–40 days before harvest, PH10–40). Red and blue indicate positive and negative associations, respectively, with color intensity representing the magnitude of the correlation coefficient. All correlation coefficients are shown; $|r| \geq 0.30$ was used as the threshold for strong associations, with $P < 0.0001$. Quality traits were assessed either at commercial harvest (fresh weight, FW; dry matter, DM) or after ripening (soluble solids content of ripe fruit, SSC_R; sugars; titratable acidity, TA; SSC_R/TA ratio; sugar/TA ratio; and vitamin C, Vc)

corresponding Spearman correlation coefficients were $r=0.38$, 0.41 , 0.46 , and 0.47 , respectively (all $P < 0.0001$), confirmed the consistent monotonic associations identified in the broader analysis. For all 4 relationships, the 95% confidence intervals around the regression lines were relatively narrow compared with the overall spread of the data points, suggesting a stable central trend despite the considerable biological variability among the accessions.

The associations between fruit quality traits and meteorological factors varied markedly across developmental

stages. Early fruit development (30–50 DAA) was the most critical window for TA and the sugar–acid balance, whereas the mid-to-late developmental stages (40–60 DAA) showed stronger associations with soluble sugar accumulation, particularly in relation to temperature variability. DM accumulation was most strongly promoted during the final preharvest period (PH20–PH40, especially PH40) by higher maximum temperatures, hot days, and a greater DTR. In contrast, higher minimum temperatures throughout the 30–60 DAA stages consistently

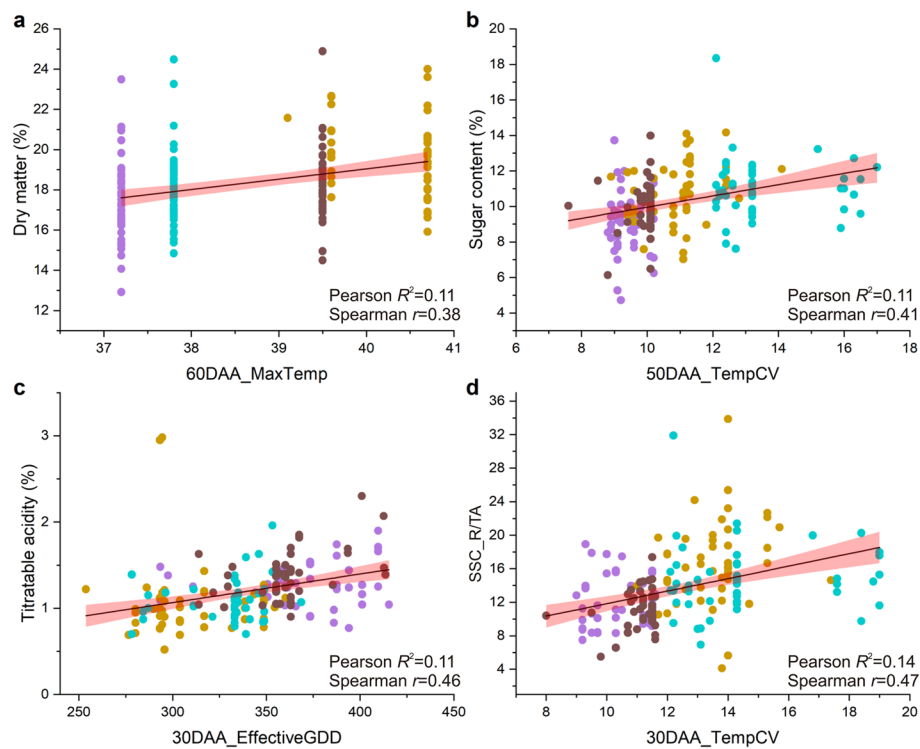


Fig. 8 Scatter plots visualizing the relationships between 4 key fruit quality traits and their most strongly correlated meteorological variables. **a** Dry matter (DM) versus maximum temperature at 60 days after anthesis (60DAA_MaxTemp). **b** Sugar content versus temperature variability at 50 DAA (50DAA_TempCV). **c** Titratable acidity (TA) versus effective growing degree days at 30 DAA (30DAA_EffectiveGDD). **d** SSC of ripe fruit (SSC_R)/TA ratio versus temperature variability at 30 DAA (30DAA_TempCV). Each point represents an individual accession sample. The solid line indicates the linear regression fit, and the shaded red band represents the 95% confidence interval of the fit

imposed a constraint on DM accumulation. Additionally, the preharvest phase was key for modulating the effects of rainfall on the acidity content.

Summarizing these correlations, the meteorological profile associated with higher kiwifruit quality (high DM and sugar, balanced acidity) likely includes: 1) adequate but not excessive early-season heat to limit high acidity; 2) a significant DTR and moderate temperature variability to promote DM and sugar accumulation; and 3) moderate preharvest rainfall to reduce acidity and improve the sugar–acid ratio. The results indicate that fruit quality is associated with multiple meteorological factors acting across distinct developmental windows rather than with a single climatic variable.

Discussion

In this study, the diversity, stability, and climatic drivers of core fruit quality traits for 47 diverse kiwifruit (*Actinidia* spp.) accessions across 4 years were systematically evaluated. Substantial phenotypic variation was observed across the germplasm collection, with Vc content and FW showing the greatest variation, whereas DM and SSC_R were comparatively stable. DM also

showed strong positive associations with SSC_R and soluble sugar content, supporting its value as a practical indicator of subsequent fruit sweetness. Genetic background was a key determinant of trait expression, and multivariate analysis grouped the accessions into 4 distinct quality profiles. Importantly, between-year variation contributed more to overall phenotypic variation than within-year variation for most quality traits, although DM, SSC_R, and several individual accessions exhibited relatively high stability. Furthermore, the climatic associations of fruit quality were strongly dependent on developmental stage, indicating that different meteorological conditions during specific phenological windows are associated with the formation of distinct quality attributes.

The genus *Actinidia* is characterized by high species diversity, comprising 54 species and 21 varieties, along with remarkable genetic variability, which results in significant differences and variation in fruit quality among different germplasms (Huang et al. 2004). Consistent with this broad genetic diversity, the present study revealed pronounced variation in Vc content, which ranged from 13.56 to 561.28 mg/100 g and exhibited the

highest CV (100.75%). Single-FW also showed considerable variation, ranging from 27.10 to 112.20 g, with a CV of 40.29%, while the DM content exhibited the smallest CV (8.87%). These results aligned with our earlier investigation of 36 kiwifruit germplasms (Shen et al. 2023). The increased CVs for Vc and single-FW can largely be attributed to the inclusion of specific germplasms: *A. rufa* typically produces small fruits with extremely low Vc content, whereas *A. eriantha* is characterized by exceptionally high Vc content and small fruit size. However, when germplasms within a more homogeneous genetic background are being examined, the CV for these traits can be substantially lower. For instance, a recent study (He et al. 2020) reported that the Vc content in 43 wild *A. eriantha* germplasms ranged from 353 to 940 mg/100 g, with a reduced CV of 19.5%. Conversely, in F_1 hybrid offspring of *A. rufa* × *A. chinensis*, the average single-FW was 32.29 g, yet the CV remained as high as 32.61%, although it was still lower than the 40.29% observed in the present study (Zheng et al. 2025). This distinction can be attributed to the composition of the materials examined herein: the 47 accessions primarily consisted of varieties or lines from *A. chinensis* var. *chinensis* and *A. chinensis* var. *deliciosa*, as well as interspecific hybrids such as *A. rufa* × *A. chinensis* and *A. chinensis* × *A. eriantha*, thereby amplifying the overall phenotypic divergence in fruit quality traits.

Correlation analysis among kiwifruit quality indicators is crucial for understanding its commercial value, post-harvest handling, and consumer preference (Titeli et al. 2023). Key quality indicators such as FW, DM, SSC_R, sugar, TA, SSC_R/TA and sugar/TA ratios, and Vc content are fundamental for assessing kiwifruit quality, and complex interrelationships exist among them (Titeli et al. 2023). In this study, consistent with previous research, significant positive correlations were found among DM, SSC_R, sugars, SSC_R/TA, and sugar/TA, whereas TA exhibited strong negative correlations with both SSC_R/TA and sugar/TA. Notably, Vc and FW showed weak or nonsignificant correlations with the major flavor-related traits, suggesting that nutritional value and fruit size represent relatively independent dimensions of fruit quality. Similar findings have been reported in a broader evaluation of 32 commercial cultivars under uniform growing conditions (Gao et al. 2026), supporting the consistency of these relationships across diverse genetic backgrounds.

Previous studies have shown that harvest DM is positively associated with SSC_R and consumer liking and can therefore serve as an important indicator of subsequent eating quality (Burdon et al. 2004; Jaeger et al. 2011). Consistent with these findings, the present study revealed strong positive relationships between harvest DM and both ripe fruit SSC and soluble sugar

content across the 47 accessions. Furthermore, the established correlation between the DM at harvest and the SSC of ripe fruits enables the development of predictive models for postharvest quality. On the basis of this relationship, a predictive model for SSC_R based on DM at harvest was proposed for the 'Hayward' cultivar: $SSC_R = -3.755 + 1.057 \times DM$ (Burdon et al. 2004). Extending this approach across genotypes, utilizing data from the 47 accessions in this study, a cross-cultivar/line generalized predictive model was also established: $SSC_R = 2.13 + 0.71 \times DM$ ($R^2 = 0.66$). Therefore, the DM content at harvest can be a robust predictor of final fruit quality and potential consumer acceptance, providing a practical tool for preharvest quality assessment and supply chain management.

In this research, 47 kiwifruit accessions were comprehensively evaluated across 8 key quality traits over 4 consecutive years. Building upon the established framework of multitrait evaluation in kiwifruit quality studies (Chen et al. 2024; Wang et al. 2025; Gao et al. 2026), we aimed to identify superior germplasms and enable precise resource characterization for targeted breeding and cultivation optimization. Using HCA and PCA, the accessions were classified into 4 distinct groups on the basis of quality traits such as FW, DM, TA, and Vc content, reflecting the genetic background more than flesh color alone. These findings differed from those of several earlier studies that employed flesh color as a primary classification criterion. For instance, Wang et al. (2025) utilized an integrated approach combining PLSR, cluster, and correlation analysis to systematically compare cultivars across different flesh colors, ultimately confirming that red-fleshed types exhibited superior overall quality and received the greatest consumer preference.

Our multiyear analysis revealed that although HCA and PCA consistently grouped accessions with similar genetic backgrounds, the annual rankings of individual accessions varied considerably. Accessions that consistently ranked high or low across years showed relatively stable performance, whereas those in the middle of the ranking exhibited greater year-to-year fluctuations. These results highlight the importance of incorporating multiple years of observations when evaluating germplasm performance. Accordingly, four-year grand mean data were used for the comprehensive quality assessment, which identified S13- 'Dongmei', S14- 'Jinmei', S18- 'Jinkui', S1- 'Mantianhong', and S9- 'Huayou' among the highest-ranked accessions. Several of these accessions also overlapped with the top-performing cultivars reported by Gao et al. (2026). In contrast, some accessions classified as low-performing in Gao et al.'s study, including 'Miliang No. 1', 'Hayward', 'Jinfeng', 'Jinyuan', and 'Jintao', showed different performance patterns in the

present evaluation, with some improving in rank while others remained relatively low-ranked. This variability underscores the significant influence of growing location and interannual environmental conditions on fruit quality, highlighting the importance of multiyear, multilocation evaluations for reliable germplasm assessment (Gao et al. 2026; Wang et al. 2025). Collectively, these findings indicate that integrating multiple years of phenotypic data can improve the reliability of elite germplasm selection by distinguishing consistently high-performing accessions from those whose quality is more variable across environments or seasons.

The stability and variability of kiwifruit quality, along with the influence of genotype $\times$ environment ($G \times E$) interactions, constitute critical considerations in breeding and cultivation management. The observed variation in quality stability among kiwifruit germplasms implies the complex interplay between genetic background and environmental conditions (Gao et al. 2026; Wang et al. 2025; Chen et al. 2024). Our stability analysis revealed that year-to-year environmental fluctuations were the predominant driver of overall variability in most quality traits, which aligns with the findings of studies on genotype–environment interactions. This finding was further corroborated by recent research confirming the significant impact of different origins and harvest years on the fruit quality of red-fleshed cultivars and ‘Hayward’ (Asadi et al. 2024; Bakoğlu et al. 2024). Moreover, numerous studies have demonstrated that cultivar (genetic background) is one of the primary determinants of fruit quality (Wang et al. 2025; Yuan et al. 2023; Zhang et al. 2020). For instance, while green- and yellow-fleshed cultivars often show greater environmental sensitivity, certain red-fleshed cultivars demonstrate more consistent trait expression across different growing regions, suggesting stronger genetic control over key quality attributes (Wang et al. 2025).

Breeding objectives typically aim to develop two types of cultivars: widely adaptable varieties that perform consistently well across diverse environments and narrowly adaptable varieties that excel under specific, optimal conditions. The identification of germplasms with both high-quality rankings and superior stability offers practical insights for cultivar deployment toward these goals. In the present study, germplasms S1, S9, S16, S18, and S43 showed relatively little year-to-year fluctuation while maintaining high quality scores, comparable to the performance stability reported for elite cultivars such as ‘Hongyang’ and ‘Jinkui’ (Gao et al. 2026). This genetic consistency is particularly valuable in breeding programs aiming to develop cultivars with reliable performance under variable growing conditions. Notably, the relatively high stability of DM and SSC_R across germplasms and years further supports their

potential as robust selection criteria. This finding is consistent with observations in wild *A. eriantha* populations, in which DM and SSC_R exhibited lower variability than other fruit-quality traits (He et al. 2020).

Kiwifruit quality is a complex trait governed by the combined effects of genotype, environment, and their interactions. The synthesis of bioactive compounds, such as anthocyanins, in red-fleshed kiwifruit is highly dependent on climatic conditions, with cooler temperatures favoring greater accumulation and color intensity (Asadi et al. 2024). In our study, some germplasms exhibited greater genetic dominance, resulting in higher cross-year stability, whereas others were more susceptible to environmental influences. The four-year experimental data in this study provide a foundation for understanding $G \times E$ interaction mechanisms and offer scientific guidance for the breeding of high-quality, widely adaptable cultivars. Nevertheless, multicenter and multi-environment trials will be necessary to validate the stability of these germplasms across different climatic and production conditions.

The substantial variation among years observed for several fruit-quality traits further indicates that environmental conditions and $G \times E$ interactions contribute importantly to quality formation. Therefore, identifying the key meteorological drivers of fruit quality formation during key development stages is essential for improving fruit quality and orchard management. However, the climatic effects identified in this study should be interpreted with appropriate caution. The linear regression models explained only a modest proportion of the variance (Pearson's $R^2 = 0.11\text{--}0.14$), indicating that climatic factors alone explain only a small fraction of the quality variation. The corresponding Spearman's rank correlation coefficients ($r = 0.38\text{--}0.47$) confirmed significant monotonic trends, but genetic background, soil properties, and orchard management practices likely played more dominant roles. The identified trends provide biologically meaningful insights into how specific meteorological factors during key developmental windows influence quality formation and should be interpreted as complementary to the well-established understanding that genetic background remains the primary determinant of fruit quality. Meteorological factors serve as pivotal drivers modulating fruit quality traits by directly influencing phenological stages and physiological and biochemical processes during development (Medda et al. 2022). These environmental variables collectively shape the ultimate qualitative composition, appearance, and marketability of plant tissues and derived products. In the broader context of climate change, such effects are of increasing concern because shifts in temperature and other climatic variables can alter kiwifruit phenology and physiology, increase

pest and disease pressure, and consequently affect yield and fruit quality (Rajan et al. 2024).

The influence of meteorological factors on fruit quality formation is complex and extends beyond the impact of any single variable. Their effects are often limited in isolation and are more likely manifested through interactions among multiple factors, with these interactions frequently dependent on specific development windows. During fruit development, specific meteorological factors interact to shape key quality attributes, such as the DM, SSC_R, sugar content, acidity, and Vc content. For example, DM accumulation appeared to be associated with a combination of temperature conditions and diurnal temperature variation. Higher maximum temperatures and a larger DTR during the 60 DAA and the period preceding harvest were associated with higher DM, potentially reflecting enhanced photosynthetic activity and net accumulation of assimilates (Woodward and Clearwater 2011). However, this positive relationship is context dependent and can be reversed under different thermal regimes. Previous research reported that elevated temperatures, specifically during the starch accumulation period (41–161 DAA), significantly reduced fruit DM in ‘Hayward’ kiwifruit (Richardson et al. 2004). Thus, temperature effects on kiwifruit quality are likely to be nonlinear and developmentally regulated, with the same meteorological factor potentially exerting contrasting effects at different stages of fruit development.

Sugar accumulation is also critically influenced by temperature patterns. A high coefficient of temperature variation during 30–60 DAA promoted sugar accumulation, whereas sustained high temperatures, indicated by mean temperature and effective GDD, tended to suppress it (Snelgar et al. 2005). A previous study revealed that maximum kiwifruit growth occurred at approximately 17 °C, with temperatures above or below this optimum reducing growth. Furthermore, warming vines during summer has been shown to reduce SSC_R (Snelgar et al. 2005). Together, these findings suggest that sugar accumulation may depend not simply on absolute temperature but also on the balance and temporal variation of thermal conditions during fruit development.

Acidity, an important determinant of sugar–acid balance and sensory quality, was associated with both early-season temperature and rainfall in the present study. Warm conditions within 30–40 DAA promoted acid accumulation, whereas rainfall and temperature fluctuations within 30–60 DAA promoted acid degradation. This pattern is broadly consistent with the findings of Richardson et al. (2004), who reported that warming during the cell division phase (0–40 DAA) increased TA, whereas warming during the starch accumulation phase

(41–161 DAA) decreased it. Although no meteorological factors were strongly correlated with the Vc content in this study, a modest negative correlation ($r=-0.29$) was observed with rainfall within 40 DAA, whereas a moderate positive correlation ($r=0.25$) was detected with 60 DAA_Eff_GDD. Richardson et al. (2004) reported that heating vines during the cell division and starch accumulation phases significantly reduced the fruit Vc content, but heating during the later maturation phase (162–194 DAA) had no significant effect. The differences between these findings may reflect variation in cultivar, environmental conditions, developmental timing, or the intensity and duration of thermal treatments, further emphasizing the complexity of environmental regulation of Vc accumulation.

Overall, optimal kiwifruit quality tends to result from a sequence of climatic interactions during key developmental stages: adequate early-season warmth helps regulate acidity, a combination of warm days and cool nights during fruit growth promotes DM and sugar accumulation, and moderate preharvest rainfall improves the sugar–acid balance. In essence, high-quality fruits depend not on isolated climate factors but on the timely orchestration of specific meteorological elements during precise developmental windows.

Conclusions

In summary, the four-year evaluation of 47 kiwifruit accessions revealed substantial phenotypic variation in fruit-quality traits. Vc exhibited the greatest variability ($CV=100.75\%$), whereas DM was the most stable trait ($CV=8.87\%$) and showed a strong positive relationship with SSC_R ($R^2=0.66$), supporting its potential as a reliable indicator of sweetness. Year-to-year environmental fluctuation was the primary driver of quality variability, while accessions S1, S9, S16, S18, and S43 exhibited both high quality and superior stability, making them elite candidates for breeding programs. Key meteorological factors during specific developmental windows showed significant monotonic associations with quality traits (Spearman’s $r=0.38$ – 0.47), although the corresponding regression models explained only partially the observed variation (Pearson’s $R^2=0.11$ – 0.14). This suggests that genetic background remains the primary determinant of fruit quality, whereas meteorological factors account for only limited variation. The potential roles of soil and management practices remain to be explored in the future. The optimal quality arises from coordinated climatic patterns—early-season warmth for acidity regulation, warm days with cool nights for DM and sugar accumulation, and moderate

preharvest rainfall for sugar–acid balance. Overall, this study identified elite kiwifruit germplasms with stable high quality and demonstrated the stage-specific associations between meteorological conditions and major quality traits. These findings provide useful resources and a scientific basis for breeding broadly adaptable, high-quality cultivars and for developing climate-adaptive strategies for kiwifruit production.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44281-026-00117-7>.

Additional file 1: Table S1. The climate information for the experimental orchard location during 2021–2024. Table S2. A comprehensive dataset of 100 derived climatic variables for 47 kiwifruit accession across key developmental windows. DAA, days after anthesis; PH10–40, pre-harvest 10–40.

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Authors' contributions

All authors contributed to the study conception and design. Plant materials were cultivated and collected by JY, QZ and FH. Quality measurement was carried out by JY, XYR, SYS, and HYL. Date collection and analysis were performed by WJH and NY. Funding acquisition was performed by CHZ. The draft manuscript was written by WJH, reviewed by CHZ, and all authors commented on previous versions of the manuscript. All authors have read and approved the final version of the manuscript.

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Data availability

Not applicable.

Declarations

Ethical approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interest

The authors declare that they have no competing interests.

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